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**Fault Detection And Bit-Error Rate Diagnostics In Optical Wireless
Links Using Edge-AI**

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Abstract

The rapid growth of Internet-of-Things (IoT) devices has placed the traditional indoor Radio Frequency (RF) spectrum under severe congestion, reducing reliable bandwidth and introducing persistent interference in dense deployment environments. Optical Wireless Communication (OWC) presents a viable alternative, utilizing the unregulated optical band to deliver high-throughput, interference-free data links. However, directed optical links are acutely sensitive to physical misalignment, ambient light interference, and transient obstructions, disruptions that cause conventional optical nodes to fail silently without any diagnostic feedback.

This project proposes an autonomous optical wireless communication testbed built on the ESP32 microcontroller. The transmitter encodes structured sensor data using On-Off Keying (OOK), while the receiver continuously monitors channel health by computing Bit-Error Rate (BER) and Received Signal in real-time. This hardware telemetry is used to train a lightweight classification model, which is then deployed directly onto the receiver node as a TinyML engine. The system classifies link conditions into fault categories.

The proposed system demonstrates a practical approach to transforming passive optical receivers into self-aware, fault-diagnosing nodes for dynamic indoor IoT environments.

Table of Contents

Abstract	i
List of Figures	iii
List of Tables	iii
Abbreviations	iv
1 Introduction	1
2 Problem Statements	2
3 Objectives	3
3.1 Main Objective	3
3.2 Specific Objectives	3
4 Literature Review	4
4.1 Review of Related Theory	4
4.1.1 Optical Wireless Communication	4
4.1.2 On-Off Keying (OOK) Modulation	4
4.1.3 Bit Error Rate and Optical Link Quality	5
4.1.4 TinyML for Edge-Based Diagnostics	5
4.2 Review of Related Works	6
4.3 Gap Analysis	6
5 Proposed Methodology	8
5.1 Research Design/Approach	8
5.2 System Architecture/Framework	8
5.3 Algorithms/Models to be Used	9
5.4 Dataset/Data Acquisition Plan	10
6 Time Schedule	11

7	Expected Outcomes	11
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List of Figures

1	<i>Proposed optical wireless communication and TinyML diagnostic architecture</i>	9
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List of Tables

1	<i>Planned Project Time Schedule Matrix</i>	11
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Abbreviations

OWC	Optical Wireless Communication
VLC	Visible Light Communication
FSO	Free Space Optics
RF	Radio Frequency
IoT	Internet of Things
OOK	On-Off Keying
BER	Bit-Error Rate
LoS	Line of Sight
ML	Machine Learning
ESP32	Espressif Systems 32-bit Microcontroller
ADC	Analog to Digital Converter

1 Introduction

Wireless communication has become a foundational requirement for modern indoor environments, driven largely by the rapid proliferation of Internet-of-Things (IoT) devices. The traditional Radio Frequency (RF) spectrum, upon which technologies such as Wi-Fi and Bluetooth depend, is increasingly congested in high-density deployments, leading to degraded throughput and persistent electromagnetic interference. Optical Wireless Communication (OWC) has emerged as a compelling alternative, utilizing the vast, unregulated optical band to deliver high-throughput, interference-free data links. Unlike RF systems, optical signals are spatially confined and do not penetrate walls, offering inherent security and enabling dense spatial reuse within the same building.

Within the broader OWC domain, two principal approaches exist. LED-based Visible Light Communication (VLC) systems leverage existing illumination infrastructure to broadcast data across a wide field, making them suitable for general-purpose indoor coverage. In contrast, directed optical sources such as laser diodes operate with near-zero beam divergence, confining transmission to a precise Line-of-Sight (LoS) path. This directional characteristic yields significantly higher attainable bit-rates, longer transmission ranges, and tighter spatial security compared to LED-based systems. However, it also introduces acute sensitivity to physical misalignment, ambient light interference, atmospheric particulates such as dust and smoke, and transient obstructions, disruptions that cause conventional optical nodes to fail abruptly without any diagnostic feedback to the user or the network.

The ESP32 microcontroller platform, featuring a dual-core Xtensa architecture with integrated analog peripherals, presents a capable and low-cost foundation for implementing optical wireless nodes. Its onboard 12-bit Analog-to-Digital Converter (ADC) and high-speed General Purpose Input/Output (GPIO) pins enable direct interfacing with optical transmitter and receiver circuitry. Furthermore, the growing availability of TinyML frameworks allows lightweight machine learning models trained on desktop platforms to be compiled and deployed directly onto such microcontrollers, enabling on-device inference without dependence on external servers or cloud connectivity.

This project proposes an autonomous optical wireless communication testbed in which the receiver ESP32 node continuously monitors channel health metrics, computed Bit-Error Rate (BER) and received signal voltage, and classifies link conditions in real-time using an edge-deployed machine learning model. The system is designed to move beyond passive data reception toward active, self-aware fault diagnostics, establishing a practical methodology for robust optical wireless communication in dynamic indoor environments.

2 Problem Statements

The increasing demand for reliable indoor wireless communication, combined with the inherent physical constraints of directed optical links, gives rise to several interconnected engineering challenges that this project seeks to address.

1. **RF Spectrum Congestion:** The traditional indoor RF spectrum, predominantly occupied by Wi-Fi (2.4 GHz and 5 GHz bands) and Bluetooth, faces critical congestion as the density of IoT devices in shared indoor environments continues to grow. This results in increased packet collisions, elevated retransmission overhead, and degraded end-to-end throughput, limitations that become particularly pronounced in environments such as hospitals, laboratories, and industrial floors where electromagnetic interference is also a safety or operational concern.
2. **Directional Sensitivity of Optical Links:** Directed optical sources feature extremely low beam divergence, demanding near-perfect Line-of-Sight (LoS) alignment between transmitter and receiver. Minor structural shifts caused by thermal expansion, vibration, or accidental contact with the mounting fixture are sufficient to cause complete signal loss. Transient obstructions such as a person crossing the beam path, dust accumulation on optical surfaces, or sudden changes in ambient illumination can equally cause the received signal voltage to drop sharply and instantaneously.
3. **Absence of Real-Time Edge Diagnostics:** Existing optical wireless systems provide no localized diagnostic capability when link failures occur. The receiver node simply stops decoding data with no information available regarding whether the failure was caused by misalignment, a physical obstruction, ambient light saturation, or hardware degradation. This absence of on-device fault diagnostics prevents prompt corrective action and limits deployment of optical wireless links in environments where link reliability is critical.

3 Objectives

3.1 Main Objective

The main objective of this project is to design and implement an autonomous optical wireless communication testbed on the ESP32 platform, integrating channel telemetry monitoring with an edge-deployed machine learning model for link fault detection and classification.

3.2 Specific Objectives

- To design and construct a directed optical wireless data link using the ESP32 microcontroller, implementing On-Off Keying (OOK) modulation.
- To develop receiver firmware that autonomously computes Bit-Error Rate (BER) and received signal voltage concurrently with data reception, streaming structured telemetry logs via serial interface.
- To train a lightweight machine learning classifier on hardware-generated telemetry data and deploy it as a TinyML engine directly onto the receiver ESP32 node.
- To demonstrate classification of link conditions into defined fault categories based on the nature of the channel disruption.

4 Literature Review

4.1 Review of Related Theory

4.1.1 Optical Wireless Communication

Optical Wireless Communication (OWC) refers to the transmission of information through free-space optical signals instead of conventional radio frequency (RF) waves. OWC systems utilize the optical spectrum, including visible light and infrared wavelengths, as the communication medium. Compared to RF-based communication, optical systems provide advantages such as immunity to electromagnetic interference, high spatial reuse, and improved security due to the directional nature of optical transmission.

Optical communication systems generally consist of a transmitter, optical channel, and receiver. The transmitter converts electrical data into variations of optical intensity using sources such as Light Emitting Diodes (LEDs) or laser diodes. The receiver uses a photodetector, commonly a photodiode, to convert the received optical signal back into an electrical form for decoding.

Free-Space Optical (FSO) communication is a form of OWC that uses a directed optical beam for point-to-point communication. While the narrow beam enables high data rates and focused transmission, it also makes the system sensitive to alignment errors, physical obstructions, and environmental noise. These factors can reduce received signal strength and introduce errors in the recovered data.

4.1.2 On-Off Keying (OOK) Modulation

On-Off Keying (OOK) is a simple intensity modulation technique widely used in optical communication systems. In OOK modulation, binary data is represented by switching the optical source between two intensity states.

A logical '1' is transmitted by turning the optical source on, while a logical '0' is represented by turning it off. Since OOK requires simple transmitter and receiver circuitry, it is suitable for low-cost optical communication implementations.

However, because information is encoded directly into optical intensity, communication quality depends strongly on the condition of the optical channel. Signal attenuation, misalignment, and external light interference can cause incorrect bit detection and increase communication errors.

4.1.3 Bit Error Rate and Optical Link Quality

Bit Error Rate (BER) is an important performance metric used to evaluate the reliability of a communication system. It is defined as the ratio of incorrectly received bits to the total number of transmitted bits during communication and can be expressed as

$$BER = \frac{N_e}{N_t} \quad (1)$$

where N_e represents the number of erroneous bits received and N_t represents the total number of transmitted bits.

In optical wireless communication systems, BER is influenced by several physical factors, including received optical signal level, signal-to-noise ratio (SNR), transmitter–receiver alignment accuracy, and environmental disturbances such as ambient light interference or physical obstructions. An increase in BER generally indicates degradation of the communication link and can therefore serve as an indicator of abnormal channel conditions.

Monitoring BER alongside other link parameters, such as the received signal level, provides valuable information about the health and performance of an optical communication channel. These measurements can be utilized to assess link quality, detect communication faults, and support the classification of different operating conditions within the optical wireless system.

4.1.4 TinyML for Edge-Based Diagnostics

Tiny Machine Learning (TinyML) is a machine learning paradigm that enables the deployment of lightweight models on resource-constrained embedded devices, such as microcontrollers, allowing data processing and inference to be performed locally without relying on cloud-based computation. This approach reduces communication overhead, improves response latency, and lowers power consumption, making it suitable for edge applications [5].

In communication systems, TinyML can be applied to analyze physical-layer parameters and detect patterns associated with channel degradation, interference, or system faults. By utilizing telemetry data such as Bit Error Rate (BER), Received Signal Strength, and other signal characteristics, lightweight machine learning models can classify operating conditions and provide real-time diagnostic insights directly on embedded devices. This enables efficient monitoring and fault detection while minimizing the need for external processing resources.

4.2 Review of Related Works

Khan and Nisar presented an overview of Visible Light Communication (VLC) systems, discussing the architecture, applications, modulation techniques, and challenges associated with optical wireless communication. The study highlighted the advantages of VLC over conventional RF communication, while also identifying practical limitations such as line-of-sight dependency, interference, and channel variability [1].

Purnita and Mondal investigated the application of machine learning techniques in DCO-OFDM based LiFi systems. Their work demonstrated that machine learning algorithms can be used to analyze optical communication parameters and improve system performance under varying channel conditions. The study shows the potential of data-driven approaches for optimization of optical wireless links [2].

Sejan et al. presented a comprehensive survey on MIMO Visible Light Communication systems, including recent developments involving machine learning techniques. The study discussed challenges such as optical channel variation, receiver alignment issues, and the requirement for intelligent adaptation mechanisms to improve communication reliability [3].

Salman et al. explored a machine learning-based approach for adaptive transmission in indoor LiFi systems. Their work utilized machine learning models for predicting transmission-related parameters and improving system performance under changing channel conditions. The study demonstrates the application of machine learning for intelligent optimization of optical communication systems [4].

The reviewed works demonstrate significant progress in optical wireless communication and machine learning-assisted optimization. However, existing studies mainly focus on improving transmission efficiency, adaptive modulation, or communication performance. Limited research has been explored on deploying lightweight machine learning models for real-time monitoring and classification of optical link conditions using physical-layer parameters such as BER and received signal characteristics.

4.3 Gap Analysis

Existing optical wireless communication research has primarily focused on developing reliable transmission techniques, improving data rates, and optimizing communication performance. Various studies have analyzed parameters such as received optical power, signal quality, and Bit Error Rate (BER) to evaluate optical link performance.

Machine learning techniques have also been introduced in optical communication systems for tasks such as channel analysis, adaptive transmission, and performance opti-

mization. However, these approaches mainly concentrate on improving communication efficiency rather than identifying the cause of link degradation.

Furthermore, conventional optical communication systems generally operate as data transmission platforms without intelligent diagnostic capability. Issues such as obstruction, alignment variation, or environmental interference can reduce communication quality, but the system typically cannot classify the underlying condition automatically.

Therefore, a gap exists in integrating physical-layer optical communication monitoring with lightweight edge-based machine learning for real-time link diagnosis. The proposed work addresses this gap by utilizing measured optical link parameters and BER-based telemetry to classify communication conditions using an embedded machine learning model.

5 Proposed Methodology

5.1 Research Design/Approach

The project will follow an experimental and iterative approach to develop and evaluate an intelligent optical wireless communication system. An indoor Visible Light Communication (VLC) link will be implemented using two ESP32-based nodes acting as transmitter and receiver.

The transmitter node will acquire sensor data from an input sensor module such as DHT11, which provides structured data such as temperature and humidity readings. The acquired sensor data will be processed by the ESP32 transmitter and encoded using On-Off Keying (OOK) modulation to drive the optical source. The transmitted optical signal will propagate through a free-space optical channel where different channel conditions will be introduced and analyzed.

The receiver node will detect the incoming optical signal using a photodetector and signal conditioning circuit. Along with recovering the transmitted data, the receiver will collect physical-layer and environmental telemetry data such as received signal characteristics, optical intensity variations, and possible environmental parameters.

The collected data will be used to train a lightweight machine learning model. The trained model will be deployed on an embedded platform to classify the operating condition of the optical link. The overall methodology consists of optical system development, data acquisition, model training, deployment, and performance evaluation.

5.2 System Architecture/Framework

The proposed system consists of three main components: an optical transmitter, a free-space optical communication channel, and a receiver node with embedded diagnostic capability.

The transmitter node (ESP32-1) receives structured sensor data from an input sensor module such as DHT11. The ESP32 processes this data and performs OOK modulation to convert the digital sensor readings into optical pulses. The modulated signal drives the laser source through a suitable driver circuit for transmission.

The optical channel represents the communication medium where practical impairments such as signal attenuation, obstruction, misalignment, and ambient light interference may affect the received signal quality.

The receiver node (ESP32-2) will use a photodetector with signal amplification and

filtering circuitry to capture and process the received optical signal. The receiver will measure relevant communication parameters and environmental telemetry using suitable sensors. These measurements will be stored and provided as input to the TinyML model for link condition classification.

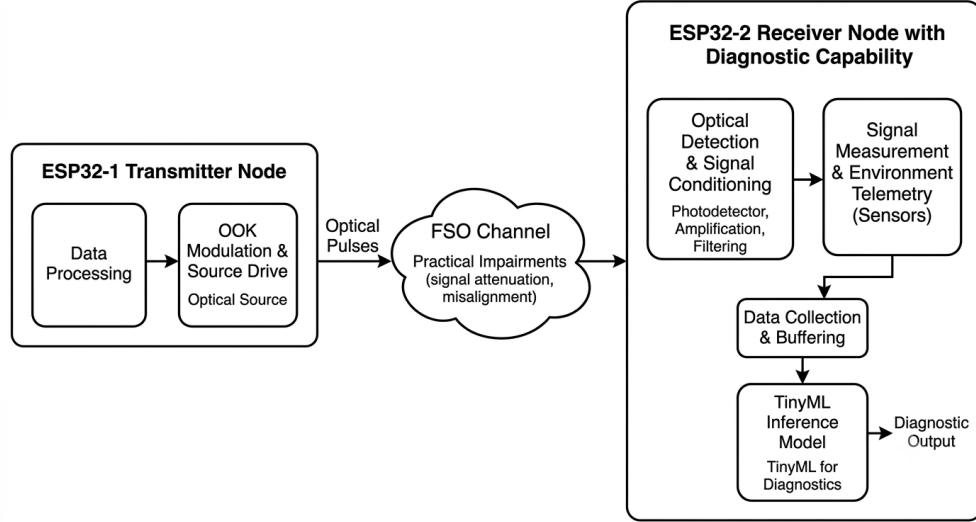


Figure 1: *Proposed optical wireless communication and TinyML diagnostic architecture*

5.3 Algorithms/Models to be Used

A supervised machine learning approach will be used to analyze the collected optical link telemetry and classify different communication conditions.

The input features for the model will be obtained from measured communication and environmental parameters. These may include:

- Received optical signal level or photodetector output
- Received signal voltage from the receiver circuit
- Bit Error Rate (BER)
- Packet transmission statistics
- Ambient environmental measurements (if available)

The target output classes will represent different operating conditions of the optical communication link. The exact classification categories will be finalized after preliminary data collection and analysis.

Possible classification outcomes may include:

- Normal communication condition
- Degraded link condition
- Signal interruption condition
- Noisy or unstable channel condition

Lightweight machine learning algorithms suitable for TinyML deployment, such as Decision Tree, Random Forest, or compact neural network models, will be evaluated. The selected model will be optimized and deployed on the embedded receiver node for real-time inference.

5.4 Dataset/Data Acquisition Plan

The dataset will be generated from the implemented optical communication system by collecting telemetry data under different operating conditions. Unlike pre-existing datasets, the collected data will represent the behavior of the developed optical link in the intended experimental environment.

The receiver node will continuously record communication parameters during operation. The collected dataset may include optical signal measurements, communication error information, and environmental parameters obtained through sensors.

The data collection process will consider different experimental conditions such as:

- Normal line-of-sight optical communication
- Variation in transmitter-receiver alignment
- Partial obstruction of the optical path
- Complete interruption of the optical path
- Changes in surrounding environmental conditions

The collected data will be stored in CSV format and processed for machine learning model development. The dataset will be divided into training, validation, and testing subsets for model evaluation. The final model performance will be measured based on classification accuracy and its ability to identify optical link conditions in real time.

6 Time Schedule

The proposed project will be completed within a period of 10 weeks. The planned activities and their timeline are shown in Table 1.

Table 1: *Planned Project Time Schedule Matrix*

S.N.	Activity / Task	W1-2	W3-4	W5-6	W7-8	W9-10
1	Hardware Interfacing & Prototype Development	•	•			
2	Sensor Data Transmission & OOK Modulation Firmware Development		•	•		
3	Optical Link Implementation & BER Logging Pipeline Design			•	•	
4	Communication Dataset Extraction & Preprocessing			•	•	
5	Supervised TinyML Model Training & Edge Model Deployment				•	•
6	System Integration, Testing & Final Evaluation				•	•
7	Report Compilation & System Review					•

7 Expected Outcomes

The proposed system is expected to demonstrate the feasibility of integrating optical wireless communication with embedded machine learning for intelligent link monitoring and diagnostics. The expected outcomes of the project are:

- A functional optical wireless communication prototype capable of transmitting sensor data between a transmitter and receiver node using an intensity-modulated optical signal.
- Successful implementation of an OOK-based communication system including sensor data acquisition, optical signal transmission, receiver-side detection, and data decoding.
- A dataset containing communication telemetry parameters such as received signal characteristics, Bit Error Rate (BER), and transmission statistics collected under different optical link conditions.

- Development of a lightweight TinyML-based classification model capable of analyzing optical link parameters and identifying different communication conditions on an embedded device.
- An edge-based diagnostic framework capable of providing real-time indication of optical link quality without requiring external computation.
- Performance evaluation of the developed system based on communication reliability, model classification performance, and response time under different experimental conditions.

The successful completion of this project will provide a low-cost prototype demonstrating the application of machine learning-based diagnostics in optical wireless communication systems.

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